**Budget Buddy** – has 3 supported Modalities:

1. Text - Natural language expense entry
2. Receipt photo upload and processing - receipt formats – jpg, png, pdf
3. Form - Traditional manual input

**Vision Processing:**

- Gemini Vision API for receipt OCR
- Text extraction → Structured parsing
- Multi-item receipt support

**Files:**

- backend/receipt_parser.py → for Vision implementation
- src/app/components/SpendingForm.tsx → UI for all 3 methods

**Example**:

- python

```
- async def parse_receipt(self, image_data: bytes):
-     image = Image.open(io.BytesIO(image_data))
-     extracted_text = await self._extract_text(image)
-     structured_data = await self._parse_structure(extracted_text)
-     return structured_data
```

**Vision Extraction** through **Gemini Vision API:**

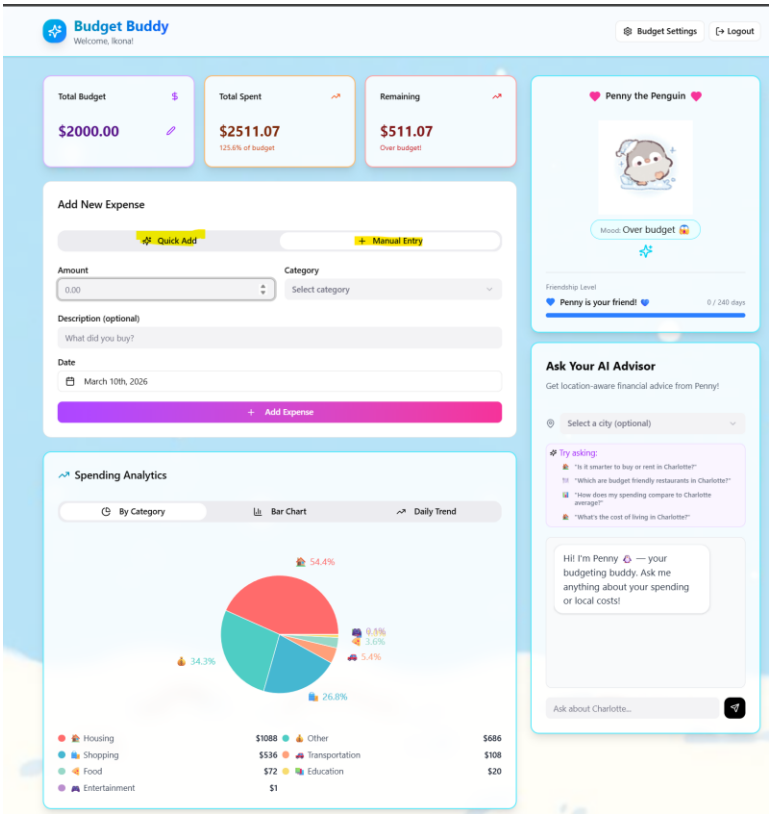
Receipt text extraction:

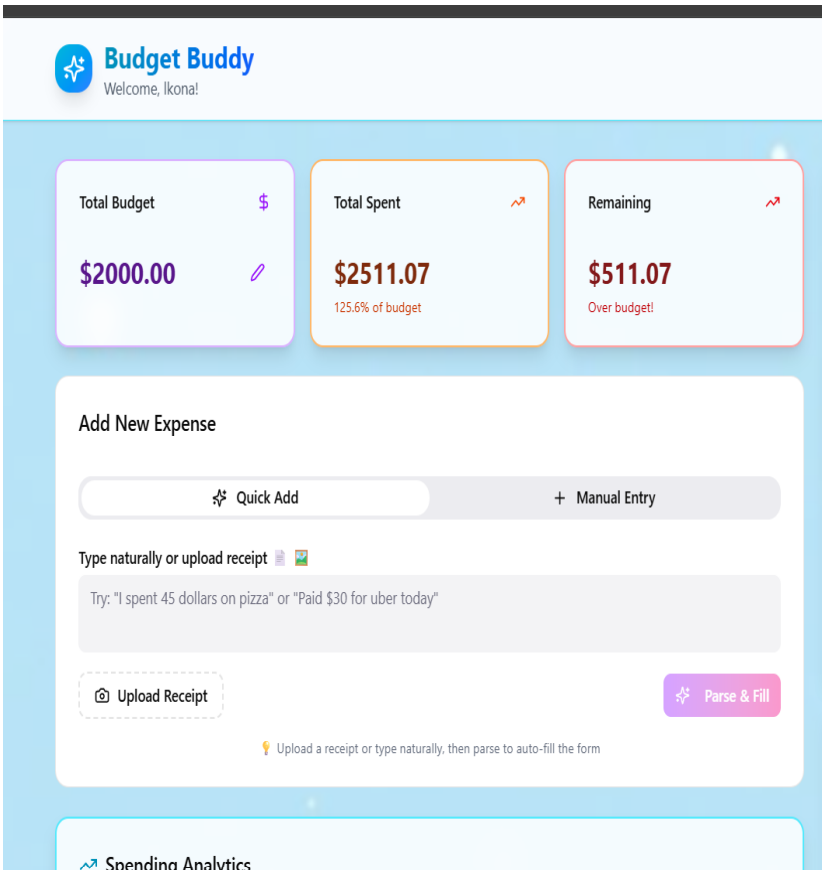
- Merchant name identification
- Item listing
- Total amount detection
- Date parsing

**Two-Stage Processing:**

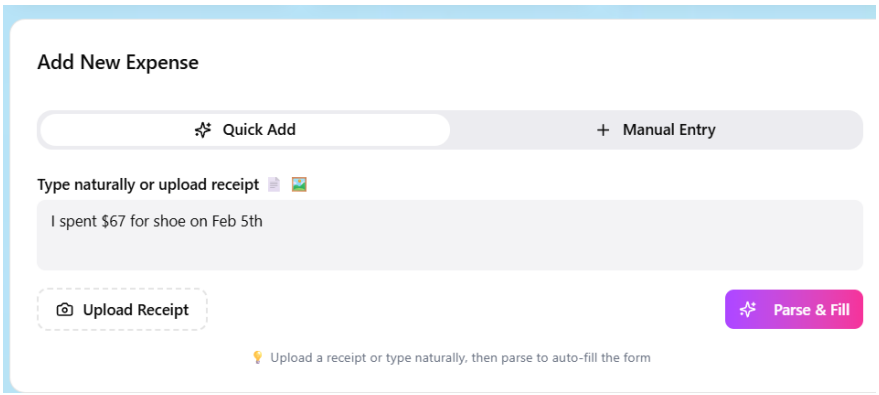
**Stage 1**: Extract all visible text from image

**Stage 2**: Parse text into structured expense data

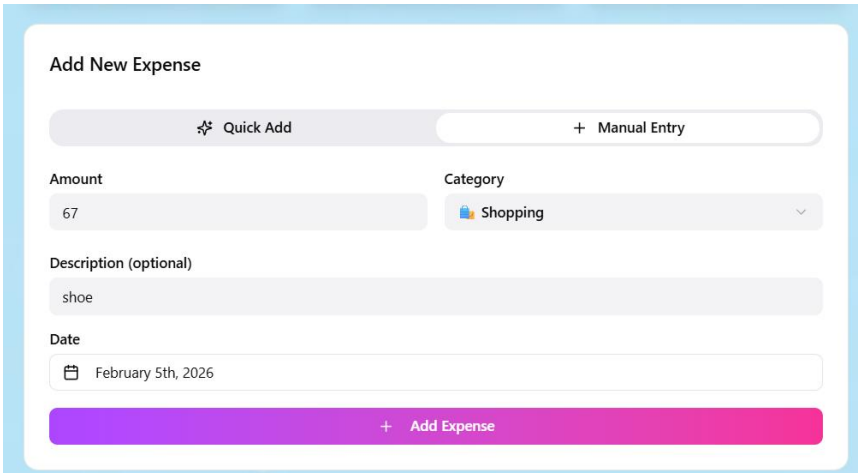

Budget Buddy
Welcome, Ikona!
Budget Settings
Logout
Total Budget
$2000.00
Total Spent
$2511.07
125.6% of budget
Remaining
$511.07
Over budget!
Penny the Penguin
Mood: Over budget
Friendship Level
Penny is your friend!
0 / 240 days
Add New Expense
Quick Add
Manual Entry
Amount
0.00
Category
Select category
Description (optional)
What did you buy?
Date
March 10th, 2026
Add Expense
Ask Your AI Advisor
Get location-aware financial advice from Penny!
Select a city (optional)
Try asking:
"Is it smarter to buy or rent in Charlotte?"
"Which are budget friendly restaurants in Charlotte?"
"How does my spending compare to Charlotte average?"
"What's the cost of living in Charlotte?"
Hi! I'm Penny — your budgeting buddy. Ask me anything about your spending or local costs!
Ask about Charlotte...
Spending Analytics
By Category
Bar Chart
Daily Trend
54.4%
34.3%
26.8%
5.4%
3.6%
Housing $1088
Shopping $536
Food $72
Entertainment $1
Other $686
Transportation $108
Education $20


Budget Buddy
Welcome, Ikona!
Total Budget
$2000.00
Total Spent
$2511.07
125.6% of budget
Remaining
$511.07
Over budget!
Add New Expense
Quick Add
Manual Entry
Type naturally or upload receipt
Try: "I spent 45 dollars on pizza" or "Paid $30 for uber today"
Upload Receipt
Parse & Fill
Upload a receipt or type naturally, then parse to auto-fill the form
Spending Analytics

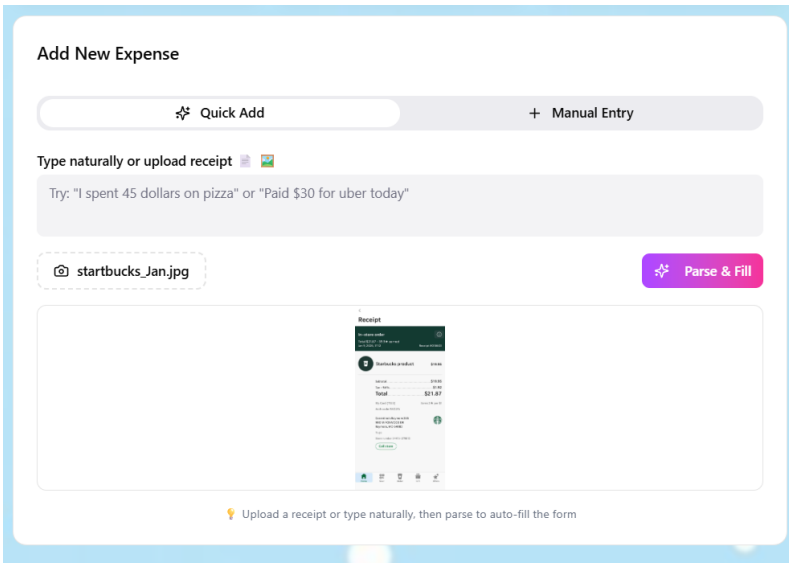
1. Text format:



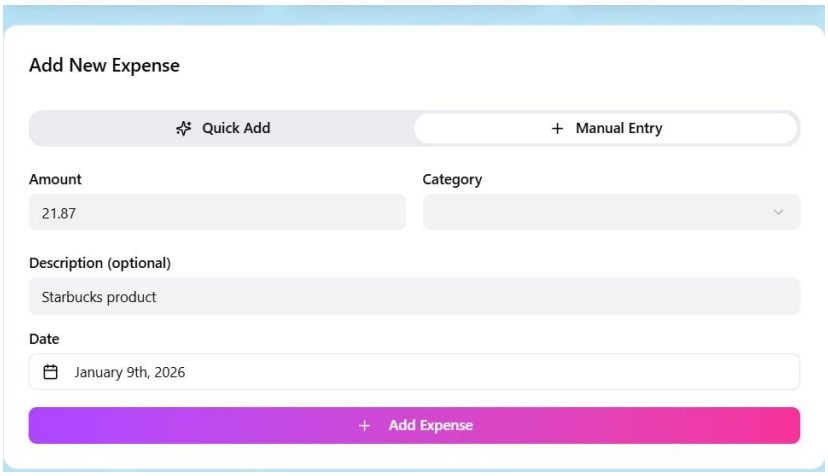

Parsed:



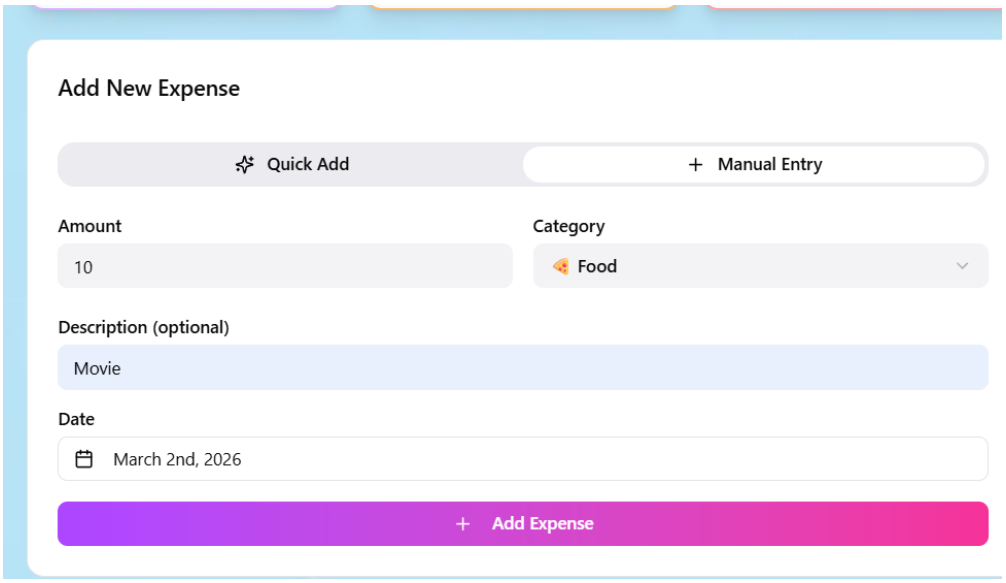

2. Upload PDF and img files



Parsed and added to backend table:




3. manual input




**Added all the records to database expense table:**

| id uuid | user_id uuid | amount nu... | category text | description text | expense_date date | created_at timestamptz | updated_at timestamptz |
|---|---|---|---|---|---|---|---|
| e0172981-e915-4d4d-830d-1ee7ed2b9da7 | 86d79cb4-0e0b-4a89-98ba-2a85... | 10.00 | Food | Movie | 2026-03-02 | 2026-03-10 21:10:24.346565+0 | 2026-03-10 21:10:24.346565+0 |
| a10f904b-3673-459a-b741-0b7f41b4c3bb | 86d79cb4-0e0b-4a89-98ba-2a85... | 21.87 | Other | Starbucks product | 2026-01-09 | 2026-03-10 21:10:01.629585+0 | 2026-03-10 21:10:01.629585+0 |
| d53ccf6e-2fe5-45ac-99dc-309c6b0ef3b6 | 86d79cb4-0e0b-4a89-98ba-2a85... | 67.00 | Shopping | shoe | 2026-02-05 | 2026-03-10 21:07:08.113355+00 | 2026-03-10 21:07:08.113355+00 |
| 44d1fdca-22f8-4df1-9ea1-814f773c27c3 | 86d79cb4-0e0b-4a89-98ba-2a85... | 3301.00 | Bills | Utility Bill | 2026-02-10 | 2026-03-10 20:48:59.350423+ | 2026-03-10 20:48:59.350423+ |
| 0e8a53b2-369e-4825-9dc7-63feba5bd52 | 86d79cb4-0e0b-4a89-98ba-2a85... | 3301.00 | Bills | Utility bill | 2026-02-10 | 2026-03-10 20:47:45.122427+0 | 2026-03-10 20:47:45.122427+0 |